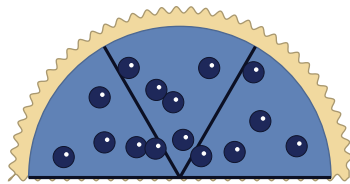


Fractions: Multiplication

Multiplying Fractions

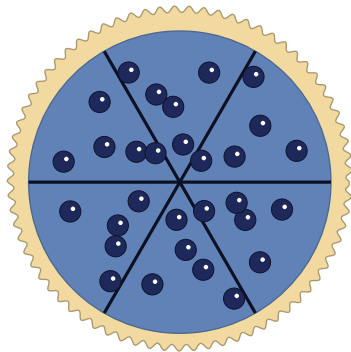
If you have half of a blueberry pie left in the fridge and eat $\frac{2}{3}$ **of** it for breakfast, what fraction of the **entire pie** did you eat for breakfast?

To eat $\frac{2}{3}$ **of** the half, you would need to divide the half into 3 equal pieces and eat 2 of them.



Alt text: A semi-circle representing the top half of a blueberry pie that is cut into three equal pieces.

If the **whole** pie was divided into slices of that same size, there would be a total of 6 slices.



Alt text: A circle representing whole blueberry pie that is cut into six equal pieces.

Since you ate 2 slices, you ate $\frac{2}{6}$ or $\frac{1}{3}$ of the **whole** pie.

In mathematics, the word **of** is code for multiplication. Therefore, to find $\frac{2}{3}$ **of** $\frac{1}{2}$, we multiply the fractions. To obtain the answer $\frac{2}{6}$ mathematically, we would have to multiply the numerators and the denominators and this is exactly how we multiply fractions.

Author(s): Kelly Stady

$$\begin{aligned}\frac{2}{3} \text{ of } \frac{1}{2} &= \frac{2}{3} \cdot \frac{1}{2} \\ &= \frac{2 \cdot 1}{3 \cdot 2} \\ &= \frac{2}{6} \\ &= \frac{1}{3}\end{aligned}$$

Procedure 1 (Multiplying Fractions). If a, b, c and d are numbers such that b and d are not equal to 0, then

$$\frac{a}{b} \cdot \frac{c}{d} = \frac{a \cdot c}{b \cdot d}$$

To multiply fractions, multiply the numerators and the denominators.

Remark 1. (a) It's a good idea to simplify fractions before multiplying. If you do so, you won't have to simplify the answer.

(b) Mixed numbers must be written as improper fractions before multiplying.

(c) When you multiply a fraction and a whole number, it can be helpful to write the whole number as a fraction with a denominator of 1. For example, if you want to find $\frac{1}{8}$ of 15, you can write 15 divided by 1 as shown below.

$$\frac{1}{8} \cdot \frac{15}{1}$$

Now, it's clear which numbers are in the numerator and denominator of each fraction.

Problem 1 Find each product. Remember to simplify your answer, if possible.

$$\frac{3}{5} \cdot \frac{4}{7} = \frac{12}{35}$$

$$\frac{12}{8} \cdot \frac{1}{4} = \frac{3}{8}$$

$$50 \cdot \frac{25}{100} = \frac{25}{2}$$

Problem 2 A cookie recipe calls for $\frac{3}{4}$ cup of flour. If you only want to make $\frac{2}{3}$ of the recipe, how many cups of flour do you need?

$$\frac{2}{3} \cdot \frac{3}{4} = \frac{1}{2} \text{ cups}$$

Problem 3 Financial advisors recommend spending no more than $\frac{3}{10}$ of your monthly take-home pay (your income after taxes and health insurance) on housing costs, like rent or a mortgage. If your monthly take-home pay is \$2500, what is the maximum amount you should pay for rent or a mortgage?

$$\frac{3}{10} \cdot 2500 = \$ 750$$

Hint: Start by writing 2500 as $\frac{2500}{1}$, then simplify before multiplying. One way is to divide 2500 by 10, then multiply the result by 3 to get the answer.